Arup Ukil . Jarvis Consulting

I'm a graduate of the University of Toronto, holding a Bachelor of Science degree in Computer Science and Mathematics. I'm a full-stack developer with practical experience from my one-year internship at SS&C Technologies, volunteer work in a small team with the charity United for Literacy, and personal projects. Most of my work and personal projects have used React/React Native for the frontend and backend frameworks such as Next.js, Spring Boot, or Django. C++ is my favourite language, and I have used it since high school for competitive programming and course projects (including a lightweight networking stack and basic computer graphics with OpenGL). I bring substantial hands-on software development experience from my undergraduate studies and early roles, and I'm looking to leverage it in an early-career software engineering position.

Skills

Proficient: React/React Native, Node.js/Next.js, HTML/CSS, TypeScript/JavaScript, C++, Java, Spring Boot, Linux/Bash, PostgreSQL/SQL, Git, Docker, Kubernetes, Spark/PySpark

Competent: Python, C, AWS, Azure, WSL, Postman, CI/CD (Jenkins), Agile/Scrum, Maven, Scala, Databricks

Familiar: Machine Learning, Python ML Libraries (NumPy, TensorFlow, PyTorch, Scikit-learn, Gensim), Flask, Django, Figma, Jira

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_ArupUkil

Cluster Monitor [GitHub]: Developed a Linux cluster monitoring agent that collects server hardware information once and then collects resource-usage metrics at fixed intervals (e.g., every minute). Tested on Rocky Linux 9 (and expected to work on similar distributions). Data is collected via Bash scripts that run standard Linux commands and is persisted in a PostgreSQL database running in a Docker container.

Grep App [GitHub]: Built a simplified clone of the Linux `grep` command that searches for regex matches across a user-provided file or directory and writes matching lines to an output file. Implemented using Core Java, Lambda/Stream APIs, SLF4J logging, and JUnit tests. Developed with Maven/IntelliJ, with deployment support via Docker for containerisation.

Stock Quote App [GitHub]: Developed a CLI Java stock trading platform that fetches real-time market data via the Alpha Vantage API and supports buying/selling positions persisted in a PostgreSQL database. Used JDBC for database access, OkHttp for HTTP requests/responses, and JUnit/Mockito/Testcontainers for testing. Emphasised clean architecture (e.g., DAO pattern and REST APIs) and used Jackson Mixins to avoid coupling the application to Alpha Vantage's response schema.

Spring Boot Trading App Backend [GitHub]: Built a Spring Boot trading system for managing traders/accounts, caching Finnhub market quotes in PostgreSQL, and submitting market orders via RESTful CRUD endpoints. Uses a layered (Controller → Service → DAO) architecture with Java 8, Spring Boot, JPA/Hibernate, and Apache HttpClient. Unit tests use Apache DBCP to provision temporary H2 DataSources, and Swagger UI is included for manual API testing.

React Trading App Frontend [GitHub]: Developed a React and TypeScript frontend for the Spring Boot trading app, including dashboard, quotes, and trader account pages for managing traders, viewing cached market data, and depositing/withdrawing funds. Used React Router for navigation, Ant Design for UI components, Axios for API calls, and Jest/React Testing Library for basic frontend testing.

Cloud, Kubernetes, and DevOps [GitHub]: Deployed the containerised Spring Boot trading app backend on Azure using Azure Kubernetes Service (AKS) and Azure Container Registry (ACR) with a fully automated CI/CD pipeline using Jenkins. The pipeline builds and pushes images onto ACR and then deploys the app for development or production.

Python Data Analytics [GitHub]: Developed a proof-of-concept analytics solution for London Gift Shop (LGS) to uncover order trends, customer retention patterns, and high-value shopper characteristics. Built the workflow using Docker, PostgreSQL, Jupyter Notebook, Python, Pandas, NumPy, Matplotlib, Seaborn, SQLAlchemy, and SQL. Then, analyzed invoice distributions, monthly KPIs, customer lifecycle metrics, and RFM segments to generate actionable business recommendations.

Spark [GitHub]: Developed Spark-based analytics workflows in Azure Databricks and Zeppelin on GCP Dataproc. Re-implemented the earlier Python Data Analytics retail notebook as a PySpark solution in Databricks using Spark SQL, PySpark DataFrames, and shared tables. Also analysed World Development Indicators data in Zeppelin using Spark SQL and PySpark for GDP-focused queries and comparisons.

Highlighted Projects

Scriptorium [GitHub]: Created and deployed a social web app on AWS (ECS, ECR, RDS) for writing, sharing, and discussing code in multiple programming languages using Next.js, React, TypeScript, and Tailwind CSS. Implemented a Prisma database schema managed via RESTful APIs with JWT-based authentication. Executed user code in isolated Docker containers to maintain security and enforce resource limits.

United for Literacy Volunteer Mobile App: Worked in a team of six for a University of Toronto partner charity, United for Literacy, to build an app that streamlines the volunteer lifecycle (application and onboarding through to offboarding). Built an in-app quiz in React Native to verify volunteers' understanding of program policies. Helped implement group chat by integrating the Django backend API with the frontend, including role-based behaviour for different user types.

Professional Experiences

Software Developer, Jarvis (2025-present): Developed technical and behavioural skills by delivering multiple projects using new tools for each. Built projects including a Linux Cluster Monitoring Agent (Bash, PostgreSQL, Docker), a Java Grep App, and a React & Spring Boot Trading App deployed on Azure Kubernetes Service. Worked in a Scrum-based process with planning, daily stand-ups, and retrospectives.

Software QA Engineer, SS&C Technologies (Sept. 2023 - Aug. 2024): Used React and Java Spring Boot to develop a test case queue manager that automates management of test runs on remote machines. Built a website navigation tool using Playwright that generates an HTML page combining a BackstopJS report with an SS&C internal files comparison report. Assisted with an Excel VBA macro to generate Progress 4GL queries for non-technical users. Migrated legacy performance scripts from LoadRunner to JMeter to reduce costs.

Research Assistant, iSE Labs (May 2022 - Aug. 2022): Developed a Python/Flask-based reverse engineering comprehension tool that parses source-code call graphs into root-to-leaf execution paths and applies agglomerative clustering to produce a hierarchical tree representation. Enhanced semantic labelling of cluster nodes using Gensim topic models (Word2Vec/Doc2Vec similarity) and TextRank for more accurate function-name-based annotations. Developed an interactive GoJS frontend to visualise comparisons of execution-path trees across software systems.

Education

University of Toronto (2020-2025), Honours Bachelor of Science, Computer Science and Mathematics

Miscellaneous

- Care Worker, Su-Rekha Care Centre (Night Shift): maintained cleanliness and safety of the care home, assisted residents as needed, and prepared supplies and common areas for the next day.
- Assistant Volunteer, Kumon: helped students from preschool to Grade 12 understand and complete Kumon math and reading assignments